

Addition and Subtraction Formulas

Formulas for sine.

$$\sin(s+t) = \sin s \cos t + \cos s \sin t$$

$$\sin(s-t) = \sin s \cos t - \cos s \sin t$$

Formulas for cosine.

$$\cos(s+t) = \cos s \cos t - \sin s \sin t$$

$$\cos(s-t) = \cos s \cos t + \sin s \sin t$$

Formulas for tangent.

$$\tan(s+t) = \frac{\tan s + \tan t}{1 - \tan s \tan t}$$

$$\tan(s-t) = \frac{\tan s - \tan t}{1 + \tan s \tan t}$$

I. Evaluating Expressions Involving Inverse Trigonometric Functions

When evaluating expressions involving inverse trigonometric functions, remember that an expression like $\cos^{-1} x$ represents a(n) angle. $\sec(\sin^{-1} \frac{1}{2} + \cos^{-1} \frac{\sqrt{3}}{2})$

$$\sin(\cos^{-1}(\frac{1}{2})) = \sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$$

II. Expressions of the Form $A \sin x + B \cos x$

We can write expressions of the form $A \sin x + B \cos x$ in terms of a single trigonometric function using the Addition Formula for Sine. If A and B are real numbers, then

$$A \sin x + B \cos x = \boxed{k \sin(x + \phi)}, \text{ where } k = \sqrt{A^2 + B^2}, \text{ and } \phi \text{ satisfies } \cos \phi = \frac{A}{\sqrt{A^2 + B^2}} = \frac{A}{k}$$

$$\text{and } \sin \phi = \frac{B}{\sqrt{A^2 + B^2}} = \frac{B}{k}$$

Example 1: Use an Addition or Subtraction Formula to find the exact value of the expression.

a. $\tan 165^\circ$

$$\begin{aligned} \tan 165^\circ &= \tan(120^\circ + 45^\circ) \\ &= \frac{\tan(120^\circ) + \tan(45^\circ)}{1 - \tan(120^\circ)\tan(45^\circ)} \\ &= \frac{-\sqrt{3} + 1}{1 + \sqrt{3}} \\ &= \frac{1 - \sqrt{3}}{1 + \sqrt{3}} \cdot \frac{1 - \sqrt{3}}{1 - \sqrt{3}} \\ &= \frac{1 - 2\sqrt{3} + 3}{1 - 3} \\ &= \frac{4 - 2\sqrt{3}}{-2} = \boxed{-2 + \sqrt{3}} \end{aligned}$$

b. $\cos \frac{17\pi}{12}$

$$\begin{aligned} \cos \frac{17\pi}{12} &= \cos \left(\frac{5\pi}{4} + \frac{\pi}{6} \right) \\ &= \cos \left(\frac{5\pi}{4} + \frac{\pi}{6} \right) \\ &= \cos \left(\frac{5\pi}{4} \cos \frac{\pi}{6} - \sin \frac{5\pi}{4} \sin \frac{\pi}{6} \right) \\ &= \cos \left(\frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} - \left(-\frac{\sqrt{2}}{2} \right) \cdot \frac{1}{2} \right) \\ &= \frac{\sqrt{6} + \sqrt{2}}{4} = \boxed{\frac{\sqrt{6} + \sqrt{2}}{4}} \end{aligned}$$

Example 2: Use an Addition or Subtraction Formula to write the expression as a trigonometric function of one number, and then find its exact value.

a. $\cos 23^\circ \cos 67^\circ - \sin 23^\circ \sin 67^\circ$

$s = 23^\circ, t = 67^\circ$

$$\begin{aligned}\cos(s+t) &= \cos s \cos t - \sin s \sin t \\ &= \cos(23^\circ + 67^\circ) \\ &= \cos(90^\circ) \\ &= \boxed{0}\end{aligned}$$

b. $\sin \frac{3\pi}{4} \cos \frac{\pi}{4} - \cos \frac{3\pi}{4} \sin \frac{\pi}{4}$

c. $\frac{\tan 55^\circ - \tan 10^\circ}{1 + \tan 55^\circ \tan 10^\circ}$
 $s = 55^\circ, t = 10^\circ$

$$\begin{aligned}\tan(s-t) &= \frac{\tan s - \tan t}{1 + \tan s \tan t} \\ &= \tan(55^\circ - 10^\circ) \\ &= \tan(45^\circ) \\ &= \boxed{1}\end{aligned}$$

d. $\cos \frac{5\pi}{18} \cos \frac{\pi}{9} + \sin \frac{5\pi}{18} \sin \frac{\pi}{9}$

Example 3: Prove the identity.

a. $\cos\left(x - \frac{\pi}{2}\right) = \sin x$

LHS: $\cos x \cos \frac{\pi}{2} + \sin x \sin \frac{\pi}{2}$

$= 0 + \sin x \cdot 1$

$= \sin x$

$= \text{RHS}$

b. $\cos\left(x + \frac{\pi}{3}\right) + \sin\left(x - \frac{\pi}{6}\right) = 0$

c. $\tan\left(x + \frac{\pi}{3}\right) = \frac{\sqrt{3} + \tan x}{1 - \sqrt{3} \tan x}$

d. $\frac{\sin(x+y) - \sin(x-y)}{\cos(x+y) + \cos(x-y)} = \tan y$

$$\begin{aligned}\text{LHS: } & \frac{\sin x \cos y + \cos x \sin y - (\sin x \cos y - \cos x \sin y)}{\cos x \cos y - \sin x \sin y + \cos x \cos y + \sin x \sin y} \\ &= \frac{\cancel{\sin x \cos y} + \cos x \sin y - \cancel{\sin x \cos y} + \cos x \sin y}{\cos x \cos y - \cancel{\sin x \sin y} + \cos x \cos y + \cancel{\sin x \sin y}} \\ &= \frac{2 \cos x \sin y}{2 \cos x \cos y} \\ &= \frac{\sin y}{\cos y} \\ &= \tan y \\ &= \text{RHS}\end{aligned}$$

$$e. \cos(x+y)\cos(x-y) = \cos^2 x - \sin^2 y$$

$$f. \csc\left(\frac{\pi}{2} - u\right) = \sec u$$

$$\text{RHS: } (\cos^2 x - \sin^2 y)$$

$$= (\cos x - \sin y)(\cos x + \sin y)$$

$$= \text{LHS}$$

$$\text{LHS: } (\cos x \cos y - \sin x \sin y)(\cos x \cos y + \sin x \sin y)$$

$$= \cos^2 x \cos^2 y - \sin^2 x \sin^2 y$$

$$= \cos^2 x (1 - \sin^2 y) - (1 - \cos^2 x) \sin^2 y$$

$$= \cos^2 x - \cancel{\cos^2 x \sin^2 y} - \sin^2 y + \cancel{\cos^2 x \sin^2 y}$$

$$= \cos^2 x - \sin^2 y$$

$$= \text{RHS}$$

Example 4: Write the given expression in terms of x and y only.

$$\frac{y}{r} = \frac{x}{1} \\ x = \sqrt{1-x^2}$$

$$a. \cos(\underbrace{\sin^{-1} x}_s - \underbrace{\tan^{-1} y}_t) \quad \frac{y}{x} = \frac{y}{1} \\ r = \sqrt{1+y^2}$$

$$b. \sin(\sin^{-1} x + \cos^{-1} y)$$

$$\cos(s-t) = \cos s \cos t + \sin s \sin t$$

$$= \left(\sqrt{1-x^2}\right)\left(\frac{1}{\sqrt{1+y^2}}\right) + x\left(\frac{y}{\sqrt{1+y^2}}\right)$$

Example 5: Find the exact value of the expression.

$$a. \sin\left(\cos^{-1}\left(\frac{1}{2}\right) + \tan^{-1} 1\right) \quad \frac{y}{x} = \frac{1}{1} \\ r = \sqrt{1+1} = \sqrt{2}$$

$$\sin(s+t)$$

$$\sin(s+t) = \sin s \cos t + \cos s \sin t$$

$$= \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right)$$

$$b. \tan\left(\sin^{-1}\left(\frac{3}{4}\right) - \cos^{-1}\left(\frac{1}{3}\right)\right)$$

$$\tan(s-t)$$

$$\tan(s-t) = \frac{\tan s - \tan t}{1 + \tan s \tan t}$$

$$\frac{y}{r} = \frac{3}{4}$$

$$x = \sqrt{7}$$

$$\frac{x}{r} = \frac{1}{3}$$

$$y = 2\sqrt{2}$$

$$= \frac{\frac{3}{\sqrt{7}} - 2\sqrt{2}}{1 + \left(\frac{3}{\sqrt{7}}\right)(2\sqrt{2})}$$

$$\frac{x}{r} = \frac{1}{2} \\ y = \sqrt{4-1} = \sqrt{3}$$

θ Q II

$\frac{y}{r} = \frac{4}{5}$
 $x = -3$

Example 6: Evaluate each expression under the given conditions.

a. $\sin(\theta - \phi)$; $\sin \theta = \frac{4}{5}$, θ in Quadrant II, $\tan \phi = \frac{1}{3}$, ϕ in Quadrant III

$$\sin(\theta - \phi) = \sin \theta \cos \phi - \cos \theta \sin \phi$$

$$= \left(\frac{4}{5}\right)\left(-\frac{3}{\sqrt{10}}\right) - \left(-\frac{3}{5}\right)\left(-\frac{1}{\sqrt{10}}\right)$$

Q III ϕ

$\frac{y}{x} = -\frac{1}{3}$
 $r = \sqrt{9+1}$
 $= \sqrt{10}$

θ Q III

$\frac{y}{x} = -\frac{4}{3}$
 $r = 5$

b. $\cos(\theta + \phi)$; $\tan \theta = \frac{4}{3}$, θ in Quadrant III, $\cos \phi = \frac{2}{3}$, ϕ in Quadrant IV

$$\cos(\theta + \phi) = \cos \theta \cos \phi - \sin \theta \sin \phi$$

$$= \left(\frac{3}{5}\right)\left(\frac{2}{3}\right) - \left(-\frac{4}{5}\right)\left(-\frac{\sqrt{5}}{3}\right)$$

ϕ Q IV

$\frac{x}{y} = \frac{2}{3}$
 $y = -\sqrt{5}$

Example 7: Write the expression in terms of sine only.

a. $-\sqrt{3}\sin x + \cos x$

b. $5(\sin 2x - \cos 2x)$

Q II

$A = -\sqrt{3}$, $B = 1$, $k = \sqrt{3+1} = 2$

$\cos \phi = \frac{A}{k} = -\frac{\sqrt{3}}{2}$, $\sin \phi = \frac{1}{2}$

$\phi = \frac{5\pi}{6}$

$$-\sqrt{3} \sin x + \cos x = 2 \sin\left(x + \frac{5\pi}{6}\right)$$